Advanced Training for Oil Tanker Cargo Operations (TA

Advanced Training for Oil Tanker Cargo Operations (TASCO)

200 Practice Questions and Answers for STCW Exams

Updated for DG Shipping India Exams | *Source: brightmariner.com*

Practice Set 1

Q1. What is the main requirement of the OPA 90?

A) New tankers and tank barges should have double hulls
B) New tankers and tank barges should not ply in US waters
C) New tankers and tank barges with single hulls can ply in us ports

Correct Answer: A (New tankers and tank barges should have double hulls)

Q2. Volume 2 of IMDG code comprises one part which is

A) Part 1 - General provisions, definitions and training
B) Part 2 - Classification
C) Part 3 - The Dangerous Goods List (DGL)
D) Part 4 - Packing and tank provisions

Correct Answer: C (Part 3 - The Dangerous Goods List (DGL))

Q3. What is the maximum limit of sulphur content in bunker fuel oil?

A) $4.5\%m/m$
B) $1.5\%m/m$
C) $3.0\%m/m$

Correct Answer: A ($4.5\%m/m$)

Q4. What is the minimum requirement for carrying garbage management plan onboard a ship?

A) Any ship above 400 GT or any ship carrying 15 persons or more
B) Any ship which is 12 metres or more in length
C) All ship, Irrespective of size and length

Correct Answer: C (All ship, Irrespective of size and length)

Q5. Petroleum liquids that have closed-cup flash points below 140 Deg F (60 Deg C) are considered as

A) Non-volatile Liquids
B) Flammable Liquids
C) Volatile Liquids
D) Combustible Liquids

Correct Answer: C (Volatile Liquids)

Q6. What does SECA stand for?

A) Sox Elimination Controlled Area
B) Sox Emission Controlled Area
C) Sox Escalation Controlled Area

Correct Answer: B (Sox Emission Controlled Area)

Q7. What are the three broad classification of cargoes transported on tankers?

A) Liquefied Natural gas, Liquefied Petroleum gas, Liquefied Nitrogen
B) Container, Dry Bulk and Refrigerated cargo
C) Petroleum liquids, chemical liquids, and special liquids
D) General cargo, Dry Break Bulk and Frozen cargo

Correct Answer: C (Petroleum liquids, chemical liquids, and special liquids)

Q8. What is the closed-cup flash point of non-volatile petroleum liquids?

A) below 140 Deg F (60 Deg C)
B) 113 Deg F (45 Deg C) and above
C) below 113 Deg F (45 Deg C)
D) 140 Deg F (60 Deg C) and above

Correct Answer: D (140 Deg F (60 Deg C) and above)

Q9. What is the Reid vapor pressure of flammable liquids of natural gasoline & Naphtha which has open cup flash point of 80 Deg F (26.7 Deg C)?

A) More than 8.5 but less than 14 psi
B) 8.5 psi and below
C) 14 psi and above
D) More than 24.6 psi

Correct Answer: A (More than 8.5 but less than 14 psi)

Q10. What is the requirement for ships to have SOPEP onboard ships?

A) SOPEP Should be carried on board tankers of 150 GT or more and on other ships of 400 GT or more
B) SOPEP should be carried onboard tankers of 150 GT or more and on all other ships irrespective of size
C) SOPEP should be carried only onboard tankers of 400 GT or more

Correct Answer: A (SOPEP Should be carried on board tankers of 150 GT or more and on other ships of 400 GT or mo

Q11. What is the meaning of a Category 2 ship?

A) Oil tanker of 5000 tones deadweight and above
B) Tanker of 20,000 deadweight and above without SBTs
C) Tanker of 20,000 deadweight and above with SBTs

Correct Answer: B (Tanker of 20,000 deadweight and above without SBTs)

Q12. What type of ships have to carry the SMPEP for NLS manual onboard their ships?

A) Ship of 400 GT or more carrying 15 persons or more
B) Ships of 150 GT or more carrying oil cargoes
C) Ships of 150 Gt or more carrying noxious liquid substances

Correct Answer: C (Ships of 150 Gt or more carrying noxious liquid substances)

Q13. Tankers carry palm oil, molasses and tallow as special liquids cargoes in bulk, which are grouped as

A) Animal/Vegetable oils
B) Miscellaneous Liquids
C) Non-volatile Liquids
D) Other chemical Liquids

Correct Answer: A (Animal/Vegetable oils)

Q14. Under which Annex is the Garbage record book required?

A) Annex II
B) Annex V
C) Annex VI

Correct Answer: B (Annex V)

Q15. Animal/Vegetable Oils carried in tankers and classified as

A) Chemical Liquid
B) Special Liquids
C) Non-volatile Liquids
D) Non-flammable Liquids

Correct Answer: B (Special Liquids)

Q16. Which part of the oil record book deals with machinery space operations?

A) Part I
B) Part II
C) Part III

Correct Answer: A (Part I)

Q17. Structure of MARPOL Convention layouts are as follows: find the correct one.

A) Convention, Protocol I, Protocol II, Protocol 1978, Protocol 1997
B) Convention, Protocol 1978, Protocol 1997, Protocol I, Protocol II
C) Protocol 1978, Protocol 1997, Protocol I, Protocol II, Convention
D) Protocol 1997, Protocol 1978, Protocol I, Protocol II, Convention

Correct Answer: B (Convention, Protocol 1978, Protocol 1997, Protocol I, Protocol II)

Q18. The IMDG Code became mandatory on

A) 1st of January 2000
B) 1st of January 2002
C) 1st of January 2004
D) 1st of January 2006

Correct Answer: C (1st of January 2004)

Q19. How should CAS be aligned with other ship's surveys?

A) Alignment with the renewal survey and no inclusion in the drydock survey
B) Alignment with the intermediate or special survey and no inclusion in the drydock survey
C) Independent of the intermediate survey and drydock survey

Correct Answer: B (Alignment with the intermediate or special survey and no inclusion in the drydock survey)

Q20. Liquids that have an open-cup flash point at or below 80 Deg F (26.7 Deg C) are classified as

A) Non-volatile Liquids
B) Flammable Liquids
C) Volatile Liquids
D) Combustible Liquids

Correct Answer: B (Flammable Liquids)

Q21. Which Annex deals with the prevention of pollution by sewage from ships?

A) Annex VI
B) Annex III
C) Annex IV

Correct Answer: C (Annex IV)

Q22. What type of pollution control is addressed by Annex II of MARPOL?

A) Pollution by oil
B) Pollution by noxious liquid substances
C) Pollution by garbage

Correct Answer: B (Pollution by noxious liquid substances)

Q23. What was the last date by which use of hydrochloroflourocarbons were prohibited?

A) 2015
B) 2010
C) 1 Jan 2020

Correct Answer: C (1 Jan 2020)

Q24. When did Annex VI of Marpol enter into force?

A) 1 July 1992
B) 19 May 2006
C) 31 December 1988

Correct Answer: B (19 May 2006)

Q25. What is the open cup flash point of the petroleum liquids that are classified as combustible liquids?

A) Above 80 Deg F (26.7 Deg C)
B) Above 60 Deg F (15.6 Deg C)
C) Below 80 Deg F (26.7 Deg C)
D) Below 60 Deg F (15.6 Deg C)

Correct Answer: A (Above 80 Deg F (26.7 Deg C))

Q26. ____ is a lowest temperature at which a substance catches fire.

A) Critical temperature
B) Ignition temperature
C) Eutectic temperature

Correct Answer: B (Ignition temperature)

Q27. What is an inert condition?

A) When the oxygen has been eliminated from a tank
B) When the oxygen is 10% by volume in a tank
C) A condition in which the oxygen content throughout the atmosphere of a tank has been reduced to 8% or less by volume by the addition of inert gas.
D) When the oxygen is 21% by volume in a tank

Correct Answer: C (A condition in which the oxygen content throughout the atmosphere of a tank has been reduced t

Q28. What factor mainly affects Vapour cloud dilution rate?

A) Quantity of Spill
B) Weather conditions
C) Thermal inversion
D) Diameter of the vapour cloud

Correct Answer: B (Weather conditions)

Q29. The substance involved in combustion is called

A) Combustible
B) Non-combustible
C) Inflammable substances

Correct Answer: A (Combustible)

Q30. What is cold work?

A) Carrying out work in a cold atmosphere
B) Work that cannot create a source of ignition
C) Carrying out work on reefer containers
D) Work that can be done only in cold temperatures

Correct Answer: B (Work that cannot create a source of ignition)

Q31. ____ burns automatically at room temperature.

A) Phosphorus
B) Sodium
C) Iron
D) Nitrogen

Correct Answer: A (Phosphorus)

Q32. Which type of anodes must not be fitted in tanks where flammable gases can be present?

A) Magnesium anodes
B) Aluminium anodes
C) Zinc anodes

Correct Answer: A (Magnesium anodes)

Q33. What is meant by "intrinsically safe"?

A) Equipment can be used in hazardous atmosphere
B) An electrical circuit is intrinsically safe if any spark produced normally or accidentally is incapable of igniting a gas mixture
C) There is no danger if sparks are produced in an unsafe atmosphere

Correct Answer: B (An electrical circuit is intrinsically safe if any spark produced normally or accidentally is incapabl

Q34. What is a hot work permit?

A) Document used for carrying out repairs under supervision
B) A document issued by a responsible person permitting specific hot work to be done during a particular time interval in a defined space
C) Document used for work in a drydock
D) Document valid for 7 days for carrying out tank cleaning

Correct Answer: B (A document issued by a responsible person permitting specific hot work to be done during a part

Q35. What is bonding?

A) Connecting together of cargo pipe line flanges
B) The connecting together of metal parts to ensure electrical continuity
C) Making a ship to shore cable connection
D) Connecting together flexible couplings on deck

Correct Answer: B (The connecting together of metal parts to ensure electrical continuity)

Q36. How Threshold Limit Values are specified?

A) PPM (Parts per Million)
B) Specific Numbers
C) Volume of Gas in Air
D) ppm/volume gas in Air

Correct Answer: A (PPM (Parts per Million))

Q37. What is pour point?

A) The temperature at which oil gets frozen
B) The lowest temperature at which a petroleum oil will remain fluid
C) The temperature at which oil cannot be discharged
D) The temperature at which oil needs to be heated

Correct Answer: B (The lowest temperature at which a petroleum oil will remain fluid)

Q38. A liquid is considered to be flammable when the flash point is less than

A) 50° F
B) 60° F
C) 70° F
D) 80° F

Correct Answer: D (80° F)

Q39. What is Upper flammable limit?

A) The limit for man entry of personnel into tanks
B) The concentration of a hydrocarbon gas in air above which there is insufficient oxygen to support combustion
C) The limit where there is insufficient inert gas to support combustion
D) The limit where there is too much inert gas to support combustion

Correct Answer: B (The concentration of a hydrocarbon gas in air above which there is insufficient oxygen to suppor

Q40. What is lower flammable limit?

A) The concentration of a hydrocarbon gas in air below which there is insufficient hydrocarbon to support combustion
B) The limit where there is insufficient oxygen to support combustion
C) The limit where there is too much oxygen to support combustion
D) The limit for man entry of personnel into tanks

Correct Answer: A (The concentration of a hydrocarbon gas in air below which there is insufficient hydrocarbon to su

Q41. What is cathodic protection?

A) Cathodes placed on the hull of a ship
B) Anodes placed in the stern of a ship
C) Protection of the hull of a ship
D) The prevention of corrosion by electrochemical techniques

Correct Answer: D (The prevention of corrosion by electrochemical techniques)

Q42. Where is smoking permitted on board ship while at a berth?

A) Designated smoking room
B) Anywhere on the ship
C) In the engine room
D) In the cargo control room

Correct Answer: A (Designated smoking room)

Q43. As a requirement of SOLAS, the number of sets of full protective clothing resistant to chemical attack that should be provided in addition to firefighters' outfits, is

A) 2 sets
B) 3 sets
C) 4 sets
D) 5 sets

Correct Answer: A (2 sets)

Q44. What is a material safety data sheet (MSDS)?

A) It is used for finding out the dangers of contract
B) It is used for finding out the effects of inhalation
C) A document identifying a substance and all its constituents
D) A document giving the chemical name of the product

Correct Answer: C (A document identifying a substance and all its constituents)

Q45. What is hot work?

A) Hot work is carried out in high ambient temperatures
B) Work involving sources of ignition or temperatures sufficiently high to cause the ignition of a flammable gas mixture.
C) The temperature is very high where the work is done
D) Work carried out using hot material

Correct Answer: B (Work involving sources of ignition or temperatures sufficiently high to cause the ignition of a flan

Q46. What is volatile petroleum?

A) Petroleum having a flashpoint above 60 Degrees C
B) Petroleum having a flashpoint above 100 Degrees C
C) Petroleum having a flashpoint below 100 Degrees C
D) Petroleum having a flashpoint below 60 Degrees C

Correct Answer: D (Petroleum having a flashpoint below 60 Degrees C)

Q47. What is flash point?

A) The lowest temperature at which a liquid gives off sufficient gas to form a flammable gas mixture near the surface of the liquid.
B) The temperature at which the gas is not flammable
C) The temperature at which the liquid stops emitting gas
D) The temperature at which an explosion is not possible

Correct Answer: A (The lowest temperature at which a liquid gives off sufficient gas to form a flammable gas mixture

Q48. What is non-volatile petroleum?

A) Petroleum having a flashpoint of 60 Degrees C or above
B) Petroleum having a flashpoint of 60 Degrees C or below
C) Petroleum having a flashpoint of 100 Degrees C or above
D) Petroleum having a flashpoint of 100 Degrees C or below

Correct Answer: A (Petroleum having a flashpoint of 60 Degrees C or above)

Q49. What is purging?

A) The introduction of inert gas into a tank already in the inert condition to further reduce the existing oxygen content or hydrocarbon gas content to a level below which combustion cannot take place if air is introduced
B) Introducing of oxygen into a tank to a high level
C) Using blowers to push oxygen into a tank
D) Using natural ventilation to gas free a tank

Correct Answer: A (The introduction of inert gas into a tank already in the inert condition to further reduce the existin

Q50. What is a Corona?

A) Corona is a diffuse discharge from a single sharp conductor that slowly releases some of the available energy. Generally, corona on its own is capable of igniting a gas.
B) Corona is a diffuse discharge from a single sharp conductor that slowly releases some of the available energy. Generally, corona on its own is incapable of igniting a gas.
C) Corona is a diffuse discharge from a single sharp conductor that slowly releases some of the available energy. Generally, corona on its own is incapable of igniting a gas if it is a lean mixture

Correct Answer: B (Corona is a diffuse discharge from a single sharp conductor that slowly releases some of the ava

Practice Set 2

Q1. What is putrifaction?

A) Most animal and vegetable oils undergo decomposition over time in stored condition
B) Most animal and vegetable oils undergo decomposition over time in stored condition, when heated
C) Most animal and vegetable oils undergo decomposition over time in stored condition, when heated beyond a particular temperature which varies for each and every oil

Correct Answer: A (Most animal and vegetable oils undergo decomposition over time in stored condition)

Q2. Fuel must be heated to its ___________ before it starts burning.

A) ignition temperature
B) flash point
C) fire point
D) boiling point

Correct Answer: A (ignition temperature)

Q3. What is Nitrogen Asphxiation?

A) It is unconsciousness caused by Nitrogen can lead to death
B) It is unconsciousness caused by Inert gas can lead to death
C) Nitrogen, which is non-flammable, non-toxic and colourless
D) Nitrogen which is undetectable to the human senses

Correct Answer: A (It is unconsciousness caused by Nitrogen can lead to death)

Q4. What is Toxicity

A) Toxicity is the degree to which a substance or mixture of substances can harm humans.
B) Toxicity means poisonous and affects you on oral consumption
C) Toxicity means Acidic
D) None of the above

Correct Answer: A (Toxicity is the degree to which a substance or mixture of substances can harm humans.)

Q5. What is Time Weighted Average?

A) Exposure time
B) Short Term exposure time
C) Highest exposure time
D) Averaging all the exposure times for a time period 8 hours

Correct Answer: D (Averaging all the exposure times for a time period 8 hours)

Q6. Which is a non-accumulator of charges?

A) Toluene is a Non-accumulator of charges
B) Cyclohexane is a Non-accumulator of charges
C) Alcohols is an Accumulator of charges
D) Kotones is a Non-accumulator of charges

Correct Answer: D (Kotones is a Non-accumulator of charges)

Q7. What is Auto-ignition?

A) The ignition of a combustible material without initiation by a spark or flame, when the material has been raised to a temperature at which self-sustaining combustion occurs.
B) Automatic ignition of material in a hot atmosphere
C) Ignition of materials when they come in contact with each other
D) Ignition caused by mixture of vapours given of by cargoes

Correct Answer: A (The ignition of a combustible material without initiation by a spark or flame, when the material ha

Q8. Rapid combustion takes place in coal mines.

A) True
B) False

Correct Answer: A (True)

Q9. Which of the following is a supporter of combustion?

A) cooking gas
B) nitrogen gas
C) oxygen gas
D) producer gas

Correct Answer: C (oxygen gas)

Q10. Sun's heat and lightning strike cause spontaneous combustion in forests.

A) True
B) False

Correct Answer: A (True)

Q11. Substances which catch fire are called acids.

A) True
B) False

Correct Answer: B (False)

Q12. Burning of candle is an example of slow oxidation.

A) True
B) False

Correct Answer: B (False)

Q13. Tank clingage depends on:

A) surface tension only
B) cohesive and adhesive forces
C) cohesive, adhesive forces and surface tension
D) cohesive forces, adhesive forces and the liquids melting point

Correct Answer: C (cohesive, adhesive forces and surface tension)

Q14. Oxygen Analyzer: what is your key finding in measuring oxygen level?

A) Portable oxygen analyzers are normally used to determine whether the atmosphere inside an enclosed space (cargo tank for example) may be considered fully inerted or safe for entry.
B) Fixed oxygen analyzers can be used for monitoring the oxygen content of the boiler uptakes and the inert gas main.
C) The common types of oxygen analyzers in use are, Paramagnetic sensors and Electrochemical sensors.
D) Oxygen is strongly paramagnetic whereas most other common gases are not.

Correct Answer: A (Portable oxygen analyzers are normally used to determine whether the atmosphere inside an enc

Q15. Key Features and Measurement of an Effective Safety Culture; identify one statement that identifies an effective safety culture

A) Recognition that all accidents are preventable and only usually occur following unsafe actions or a failure to follow established procedures
B) Individual seafarers assume responsibility for safety rather than relying on others to provide it
C) Through mutual respect, increasing confidence in the value of the safety culture results in a more effective Safety Management System

D) A safety management system that is always setting targets for continuous improvement, with a goal of zero accidents

Correct Answer: D (A safety management system that is always setting targets for continuous improvement, with a g

Q16. What is the maximum interval of time between sampling and analysing for each sampling head location sequentially?

A) Not exceeding 50 minutes interval
B) Not exceeding 40 minutes interval
C) Not exceeding 30 minutes interval
D) Not exceeding 60 minutes interval

Correct Answer: C (Not exceeding 30 minutes interval)

Q17. For entering an enclosed space, 21% ___________ is necessary.

A) Oxygen
B) carbondioxide
C) Nitrogen

Correct Answer: A (Oxygen)

Q18. What is the least a person should do before entering machinery spaces?

A) Wear appropriate personal protective equipment
B) Inform the bridge
C) Inform his superiors

Correct Answer: A (Wear appropriate personal protective equipment)

Q19. Class D fires involve

A) Solid cellulosic materials such as wood, paper, clothing, etc.
B) Vapour/air mixture over the surface of flammable liquids
C) Energised electrical equipment
D) Combustible materials such as magnesium, sodium, etc.

Correct Answer: D (Combustible materials such as magnesium, sodium, etc.)

Q20. The hydrocarbon content in the atmosphere of a pump room is measured using a

A) Tankscope
B) Explosimeter
C) The Draeger instrument
D) Alarm system detector

Correct Answer: C (The Draeger instrument)

Q21. What is the limitation of a combustible gas detector of catalytic combustion type?

A) It cannot be used for measurement in air
B) It cannot be used for measurement in nitrogen
C) It cannot be used for measurement in methane
D) It cannot be used for measurement in oxygen

Correct Answer: B (It cannot be used for measurement in nitrogen)

Q22. The manifold pressure when loading oil cargo should be

A) Above the cargo tank pressure
B) Below the cargo tank pressure
C) Above atmospheric pressure
D) 'Zero'

Correct Answer: D ('Zero')

Q23. What is the maximum reliable shelf life of a filter gas mask canister if the seal is unbroken?

A) 1 year from the date of manufacture
B) 3 years from the date of manufacture
C) 5 years from the date of manufacture
D) 7 years from the date of manufacture

Correct Answer: C (5 years from the date of manufacture)

Q24. Vapours having low relative density than air will be found at the ____________ of the enclosed space.

A) middle
B) top
C) bottom

Correct Answer: B (top)

Q25. Which step is NOT generally taken when gas-freeing a tank?

A) Washing the tank interior with sea water
B) Application of degreasing solvents
C) Removal of corrosion products and sludge
D) Fresh air ventilation

Correct Answer: B (Application of degreasing solvents)

Q26. The reading of a combustible gas indicator indicates the percentage of the

A) lower explosive limit of a flammable gas concentration
B) upper explosive limit of a flammable gas concentration
C) concentration of flammable gas in a compartment
D) concentration by weight of nonflammable gas in a compartment

Correct Answer: A (lower explosive limit of a flammable gas concentration)

Q27. A crew member suffering from hypothermia should be given ___________ .

A) small dose of alcohol
B) treatment for shock
C) a large meal
D) a brisk rub down

Correct Answer: B (treatment for shock)

Q28. What instrument is used to measure hydrocarbon gas in an inert atmospheres?

A) Explosimeter
B) The Draeger instrument
C) The Multi-Gas detector
D) Tankscope

Correct Answer: D (Tankscope)

Q29. Operating Principle of the Infrared instrument: identify the key principle that leads to detection of hydrocarbon gas,

A) The infra-red (IR) sensor is a transducer for the measurement of the concentration of hydrocarbons by the absorption of infra-red radiation.
B) The vapor to be monitored reaches the measuring chamber by diffusion or by means of a pump.
C) An infrared source illuminates a volume of gas that has entered inside the measurement chamber.
D) The gas absorbs some of the infrared wavelengths as the light passes through it, while others pass through it completely unattenuated.

Correct Answer: D (The gas absorbs some of the infrared wavelengths as the light passes through it, while others pa

Q30. Areas like cargo residues, cargo and tank coatings must be suspected for oxygen-deficient areas.

A) True
B) False

Correct Answer: A (True)

Q31. What should be prohibited in the vicinity of battery rooms?

A) Painting
B) Washing
C) Smoking

Correct Answer: C (Smoking)

Q32. Safe ship operations include:

A) Running the ship profitably
B) Deliver cargo safely
C) No damage to the environment

Correct Answer: C (No damage to the environment)

Q33. Which of the following current level leads to muscular contraction?

A) 0.01A
B) 0.015A
C) 0.5A
D) 0.10A

Correct Answer: B (0.015A)

Q34. Is an enclosed space entry permit required for entering pump room?

A) The Master will decide based on findings by Chief Officer
B) The chief officer will assess the situation and decide
C) Yes, the checklist, permit and other procedures apply for all enclosed spaces
D) No, the pumproom has a fixed fan for ventilation and frequently visited

Correct Answer: C (Yes, the checklist, permit and other procedures apply for all enclosed spaces)

Q35. What should you do when the alarm bell on a self-contained breathing apparatus sounds?

A) Immediately evacuate the contaminated area.
B) Open the bypass valve on the regulator and immediately evacuate the contaminated area.
C) Move the tank selector lever to the full tank position and reset the alarm so you can evacuate the area when it sounds again.
D) Move the reserve lever to the 'reserve' position on the regulator and reset the alarm so you can evacuate the area when it sounds again.

Correct Answer: A (Immediately evacuate the contaminated area.)

Q36. Experience indicates that hazardous potentials in respect of static electricity do not occur if the velocity of liquid, while loading is: Find the correct answer

A) Below $7~m/s$
B) Above $7~m/s$
C) Above $10~m/s$
D) Below $3~m/s$

Correct Answer: A (Below $7~m/s$)

Q37. Electrochemical Sensors: identify a key finding for its use on board tankers

A) Electrochemical sensors are based on the fact that cells can be constructed that react with the measured gas and generate an electric current.
B) The Cell current can be measured, and the amount of gas determined.
C) The sensors are low cost and are small enough to allow several to be incorporated into the same instrument, making them suitable for use in multi-gas detectors.
D) There are numerous electrochemical sensors available covering a number of gases which may be present in the shipboard environment.

Correct Answer: C (The sensors are low cost and are small enough to allow several to be incorporated into the same

Q38. Which of the following conditions represents a particular advantage of using a positive pressure type self-contained breathing apparatus in an atmosphere that is immediately dangerous to life or health?

A) The equipment is compact and the wearer can work in confined spaces without difficulty.
B) The equipment used is lightweight and easy to wear by reducing physical strain on the wearer.
C) The average operating time for most air cylinders is over an hour.
D) The positive pressure in the face piece prevents contaminated air from entering the face piece.

Correct Answer: D (The positive pressure in the face piece prevents contaminated air from entering the face piece.)

Q39. Which of the following is an entry activity?

A) Permit closed
B) Confirmation of removal of all equipment
C) Entry permit issued

Correct Answer: C (Entry permit issued)

Q40. Which of the following is a particular challenge when entering fuel oil tanks?

A) slippery surface
B) corrosion
C) low lighting
D) sediments

Correct Answer: A (slippery surface)

Q41. Water extinguishes a fire by

A) Heat removal or cooling
B) Smothering or oxygen exclusion
C) Flame inhibition

Correct Answer: A (Heat removal or cooling)

Q42. What is an EEBD used for?

A) For escape from machinery or accommodation spaces in the event of a fire
B) For safety when entering a tank
C) For rescuing a person from a tank
D) For supplying oxygen to a victim

Correct Answer: A (For escape from machinery or accommodation spaces in the event of a fire)

Q43. To detect the presence of explosive gases in any space, tank, or compartment, you should use a

A) flame scanner
B) halide torch
C) combustible gas indicator
D) detector filament

Correct Answer: C (combustible gas indicator)

Q44. A man has suffered a burn on the arm. There is extensive damage to the skin with charring present. How is this injury classified using standard medical terminology?

A) Dermal burn
B) Third-degree burn
C) Major burn
D) Lethal burn

Correct Answer: B (Third-degree burn)

Q45. Which of the following is required to be included in the fireman's (emergency) outfit?

A) Chemical protection face shield
B) Approved work vest
C) Self-contained breathing apparatus
D) 5 cell approved flashlight

Correct Answer: C (Self-contained breathing apparatus)

Q46. What is the least you must do before entering a cofferdam in the engine room?

A) Open up the manhole and enter the cofferdam
B) Complete an enclosed space checklist
C) Enter the cofferdam and simultaneously start ventilation

Correct Answer: B (Complete an enclosed space checklist)

Q47. Which of the following activity is done on completion of work?

A) Closing and securing all openings
B) Satisfactory lighting
C) Intimating all the related parties about the intended entry

Correct Answer: A (Closing and securing all openings)

Q48. Carbon dioxide extinguishes a fire by

A) Heat removal or cooling
B) Smothering or oxygen exclusion
C) Flame inhibition

Correct Answer: B (Smothering or oxygen exclusion)

Q49. When the percentage oxygen is between ____________ one may feel, fainting, nausea vomiting.

A) 19.5% and 16%
B) 16% and 12%
C) 12% and 10%
D) 10% and 8%

Correct Answer: D (10% and 8%)

Q50. If there has been a fire in a closed unventilated compartment it may be unsafe to enter because of

A) excess hydrogen
B) unburned carbon particles
C) excess nitrogen
D) a lack of oxygen

Correct Answer: D (a lack of oxygen)

Practice Set 3

Q1. Which are the international bodies that provide regulatory standards for electrical equipment certification?

A) IEEC and CENELEC
B) IEC and CENELEC
C) IEEC and ISO
D) IMO and ISO

Correct Answer: B (IEC and CENELEC)

Q2. Why is the 'maximum surface temperature' identified, for electrical equipment used in hazardous areas onboard LNG carriers?

A) The limit is on minimum surface temperature on LNG carriers as the cargo is cryogenic
B) Equipment should operate within these limits to reduce the power and boil-off consumption
C) Equipment should operate within these limits to prevent the formation of a hot surface
D) Equipment should operate within these limits to reduce the heat transfer to the cryogenic cargo

Correct Answer: C (Equipment should operate within these limits to prevent the formation of a hot surface)

Q3. How is intrinsic safety achieved for electrical circuits used in hazardous areas?

A) By limiting the ignition energy and surface temperature that can arise during operation
B) By limiting the number of equipment used in a space
C) By increasing mechanical ventilation to the area
D) By installing machinery which have higher power and thus shorter operation time

Correct Answer: A (By limiting the ignition energy and surface temperature that can arise during operation)

Q4. Where can electrical equipment be classified as exit are used?

A) In Zones 0, 1 and 2
B) In Zones 1 and 2
C) In Zones 0 and 2
D) In Zones 0 and 1

Correct Answer: B (In Zones 1 and 2)

Q5. Pick the IEC symbol for explosion proof electrical protection?

A) ex-e
B) ex-n
C) ex-d
D) ex-p

Correct Answer: C (ex-d)

Q6. In cleaning up an oil spill, straw is an example of a

A) chemical agent
B) blocker
C) sorbent

Correct Answer: C (sorbent)

Q7. Heavy fuel oils when spilled are

A) more harmful to sea life than lighter oils
B) easier to clean up than lighter refined oils
C) less harmful to sea life than lighter oils
D) not a real threat to marine life

Correct Answer: C (less harmful to sea life than lighter oils)

Q8. For a vessel approaching a port of reception, what the vessel must check before the transfer of residues from the ship?

A) Ports and terminals must provide reception facilities
B) Ports and terminals must have Certificates of Adequacy
C) Ports must Apply for a Certificate of Adequacy.
D) Port Reception facility operations procedures

Correct Answer: B (Ports and terminals must have Certificates of Adequacy)

Q9. Which one of the following discharges have the most long-term ecological impact due to pollution?

A) Water ballast in oil fuel tanks.
B) Ballast added to cargo tanks.
C) Discharges of clean and segregated ballast
D) Machinery space bilges.

Correct Answer: A (Water ballast in oil fuel tanks.)

Q10. What assigns for each new ship a calculable figure that will denote its total emissions of CO2 from combustion of fuel?

A) Energy Efficiency Design Index
B) Energy Inefficiency Design Index
C) SOX Design Index
D) VOX Design Index

Correct Answer: A (Energy Efficiency Design Index)

Q11. When did Annex VI take effect?

A) May 19, 2005
B) October 12, 1991
C) February 25, 1997
D) July 3, 2015

Correct Answer: A (May 19, 2005)

Q12. The extent of MARPOL compliance is determined by:

A) Vessel type, size and speed
B) Vessel flag, type and age
C) Vessel type, size and crew
D) Vessel type, size and age

Correct Answer: D (Vessel type, size and age)

Q13. MARPOL applies to:

A) applies to all ships, oil platforms, barges and private yachts
B) tankers greater than 150GT and cargo ships greater than 400GT
C) applies to all ships, oil platforms, barges and private yachts and hydrofoils
D) applies to all ships, oil platforms, barges and private yachts and hovercrafts

Correct Answer: A (applies to all ships, oil platforms, barges and private yachts)

Q14. When you have completed bunkering operations, the hoses should be ____________ .

A) blown down with inert gas
B) drained into drip pans or tanks
C) stowed with their ends open for venting
D) steam cleaned and flushed with hot water

Correct Answer: B (drained into drip pans or tanks)

Q15. What is the most important regulatory requirement for the master arriving at the lightering position?

A) Lightering of Oil and Hazardous Material Cargoes Pre-arrival notices.
B) Reporting of incidents.
C) Designation of lightering zones.
D) Factors considered in designating lightering zones.

Correct Answer: A (Lightering of Oil and Hazardous Material Cargoes Pre-arrival notices.)

Q16. Most minor spills of oil products are caused by

A) equipment failure
B) human error
C) major casualties
D) unforeseeable circumstances

Correct Answer: B (human error)

Q17. Containment and Damage Control: identify which step is of the highest importance for personal protection?

A) Don required PPE as determined from the MSDS.
B) Fight fire (if any), being careful to use firefighting methods compatible with the material involved.
C) Shut off or otherwise stem the spill at the source.
D) Contain liquid material using barriers, such as absorbents, rags or other equipment suitable to dam the flow.

Correct Answer: A (Don required PPE as determined from the MSDS.)

Q18. Segregated ballast tanks are required to be fitted on

A) Every crude oil and new product tanker of 20,000dwt and above
B) Every crude oil tanker of 20,000dwt and above and every new product tanker of 30,000dwt and above
C) Every crude oil and new product tanker of 30,000dwt and above

Correct Answer: B (Every crude oil tanker of 20,000dwt and above and every new product tanker of 30,000dwt and ab

Q19. What is used to incinerate solid and liquid wastes arising from the operation of the ship?

A) Incinerator
B) Oily water separation
C) ISM Screen
D) Fire Sprinkler Cleaner

Correct Answer: A (Incinerator)

Q20. The Volatile Organic Compound (cargo vapor) Management Plan is required on what type of vessel?

A) Tank sensors
B) Container vessels
C) Ro-ro vessels
D) Crude Oil Carriers

Correct Answer: D (Crude Oil Carriers)

Q21. Your vessel is carrying 24,000 barrels (3816 cubic metres) of oil for discharge. The cargo hoses have an inside diameter of eight inches. The container around the loading manifold must hold

A) three barrels
B) four barrels
C) six barrels
D) eight barrels

Correct Answer: A (three barrels)

Q22. It is generally NOT allowed to clean up an oil spill by using ___________ .

A) boom
B) suction equipment
C) chemical agents
D) skimmers

Correct Answer: C (chemical agents)

Q23. The total quantity of oil discharged at sea on new tankers must not exceed:

A) 1/15000 of the total quantity of that particular cargo of which the residue formed part
B) 1/2000 of the total quantity of that particular cargo of which the residue formed part
C) 1/25000 of the total quantity of that particular cargo of which the residue formed part
D) 1/30000 of the total quantity of that particular cargo of which the residue formed part

Correct Answer: D (1/30000 of the total quantity of that particular cargo of which the residue formed part)

Q24. A starting interlock in the ODME ensures

A) the monitoring equipment is fully operational before commencing the discharge of the oil effluent at sea
B) prevents malfunctioning of the system
C) that an alarm is given off when the system malfunctions
D) stops discharge of the oil effluent when the oil content exceeds 5 part per million

Correct Answer: A (the monitoring equipment is fully operational before commencing the discharge of the oil effluent

Q25. A gas venting system on a tanker that has gas lines in each tank connected to a common gas main running the full length of the ship is called:

A) gas-main system
B) A common-main system
C) A vent-main system
D) An independent system

Correct Answer: B (A common-main system)

Q26. The inert gas is delivered to the cargo tanks by an inert gas deck main line. Find the correct labelling for the deck equipment items.

A) 1-Pressure/Vacuum Valve; 2- Cargo tank valves; 3- Mast Riser; 4-Isolating Valve
B) 1-Mast Riser; 2-Isolating Valve; 3-Cargo tank valves; 4- Pressure/Vacuum Valve
C) 1-Cargo tank valves; 2- Pressure/Vacuum Valve; 3- Mast Riser; 4-Isolating Valve
D) 1-Isolating Valve; 2- Pressure/Vacuum Valve; 3- Mast Riser; 4-Cargo tank valves

Correct Answer: C (1-Cargo tank valves; 2- Pressure/Vacuum Valve; 3- Mast Riser; 4-Isolating Valve)

Q27. LNG liquid pressure is the function of liquid density and liquid level.

A) True
B) False

Correct Answer: A (True)

Q28. Pneumatically Operating Valve; what is the most critical component for correct valve operation, based on your shipboard experience of running such a system?

A) Valve orifice
B) Valve stem
C) The Diaphragm
D) The valve springs

Correct Answer: C (The Diaphragm)

Q29. Advantages of using inert gas, pick one that is the main reason for its application,

A) No explosive mixtures can form in the tank.
B) Reduces corrosion.
C) Voyage cleaning of tanks is unnecessary.
D) Reduces pumping time because of positive pressure in the tanks.

Correct Answer: A (No explosive mixtures can form in the tank.)

Q30. Which statement about a centrifugal cargo pump is TRUE?

A) It is a positive displacement pump.
B) It must have a positive suction.
C) Increasing rotation speed will decrease discharge pressure.

Correct Answer: B (It must have a positive suction.)

Q31. The total capacity of the slop tank or tanks should be not less than what percentage of the oil-carrying capacity of the vessel?

A) 3%
B) 5%
C) 7%
D) 10%

Correct Answer: A (3%)

Q32. Acidic solutions are formed in cargo tanks of tankers

A) when sulphur compounds in oil cargoes react with water
B) when sulphur compounds in oil cargoes react with oxygen
C) when sulphur compounds in oil cargoes react with hydrocarbon gas
D) when sulphur compounds in oil cargoes react with chemicals

Correct Answer: A (when sulphur compounds in oil cargoes react with water)

Q33. Tank level measurement of free water by gauging tape and plumb bob: find the labeling combination that is correct,

A) 1 Water Innage; 2-Water Cut; 3- Water Cut; 4- Water Ullage
B) 1- Water Cut; 2- Water Innage; 3- Water Cut; 4- Water Ullage
C) 1- Water Cut; 2- Water Innage; 3- Water Ullage; 4- Water Cut
D) 1- Water Cut; 2- Water Cut; 3- Water Innage; 4- Water Ullage

Correct Answer: B (1- Water Cut; 2- Water Innage; 3- Water Cut; 4- Water Ullage)

Q34. An inert gas generator burns what oil to generate inert gas?

A) Crude oil
B) Heavy oil
C) Diesel oil
D) Gas oil

Correct Answer: D (Gas oil)

Q35. Float gauges can only record liquid level greater than ____________ inches.

A) Four
B) five
C) ten

Correct Answer: A (Four)

Q36. Back flow of inert gas is prevented by:

A) An isolating valve
B) The deck water seal
C) A valve provided in the scrubber
D) The deck mechanical non-return valve

Correct Answer: B (The deck water seal)

Q37. Continuous exposure of the float to the fluctuation in sea level can damage the ___________

A) Tape tensioning device
B) diaphragm
C) sensors

Correct Answer: A (Tape tensioning device)

Q38. The float gauge is provided with a ___________ valve.

A) Ball
B) gate
C) pressure

Correct Answer: B (gate)

Q39. Tank coating requirements developed by IMO are applicable to

A) new crude oil tankers of 5000 GRT and above
B) new crude oil tankers of 5000 DWT and above
C) new crude oil tankers of 500 GRT and above
D) new crude oil tankers of 500 DWT and above

Correct Answer: B (new crude oil tankers of 5000 DWT and above)

Q40. A deepwell pump is a type of ___________ .

A) screw pump
B) centrifugal pump
C) eductor
D) gear pump

Correct Answer: B (centrifugal pump)

Q41. Bottom washing:

A) is done to remove traces of the previous cargo
B) will remove heavy wax sediments at the tank bottom
C) is not effective when carrying a small quantity of refined products
D) cannot be done with chemical solvents

Correct Answer: A (is done to remove traces of the previous cargo)

Q42. Float gauge is a commonly used device in most of the tankers.

A) True
B) False

Correct Answer: A (True)

Q43. An inert gas system is required to be fitted:

A) On all tankers of above 20,000 tonnes deadweight
B) On product tankers of above 20,000 tonnes deadweight
C) On crude oil tankers of above 20,000 tonnes deadweight
D) On chemical tankers of above 20,000 tonnes deadweight

Correct Answer: C (On crude oil tankers of above 20,000 tonnes deadweight)

Q44. Radar gauge is a type of tank gauging equipment.

A) True
B) False

Correct Answer: A (True)

Q45. Centrifugal pumps are otherwise known as

A) Reciprocating pumps
B) Gravity pumps
C) Positive displacement pumps
D) Hydro-dynamic pumps

Correct Answer: D (Hydro-dynamic pumps)

Q46. The ring main pipeline system is extensively used on:

A) VLCCs
B) Crude oil tankers
C) Medium range tankers
D) Product carriers

Correct Answer: D (Product carriers)

Q47. Class B fires are caused by

A) combustible solids
B) flammable gases
C) flammable liquids
D) electricity

Correct Answer: C (flammable liquids)

Q48. Use of Pressure-Vacuum (PV) Relief Valve; label the diagram correctly:

A) 1-IG Main; 2-Deck Isolating Valve; 3- Vent Main; 4- By-Pass Valve; 5- PV valve; 6- Tank Isolating Valve
B) 1-Deck Isolating Valve; 2- Vent Main; 3- IG Main; 4- By-Pass Valve; 5- PV valve; 6- Tank Isolating Valve
C) 1-Deck Isolating Valve; 2- IG Main; 3- Vent Main; 4- By-Pass Valve; 5- PV valve; 6- Tank Isolating Valve
D) 1-Deck Isolating Valve; 2- IG Main; 3- Vent Main; 4- By-Pass Valve; 5- Tank Isolating Valve; 6- PV valve

Correct Answer: C (1-Deck Isolating Valve; 2- IG Main; 3- Vent Main; 4- By-Pass Valve; 5- PV valve; 6- Tank Isolating V

Q49. Cargo tank fill alarm indicator panels:

A) are normally fitted on the bridge and sometimes on deck
B) are normally fitted in the engine control room and sometimes on deck
C) are normally fitted on the bridge and sometimes in the engine control room
D) are normally fitted in the cargo control room and sometimes on deck

Correct Answer: D (are normally fitted in the cargo control room and sometimes on deck)

Q50. Identify the equipment in the IG system correctly:

A) 1-O2 Analyzer; 2-Demister; 3-Air Intake-Gas Freeing; 4- Regulating Valves; 5- Blowers
B) 1- Demister; 2-O2 Analyzer; 3-Air Intake-Gas Freeing; 4- Regulating Valves; 5- Blowers
C) 1- Demister; 2-O2 Analyzer; 3 - Blowers; 4- Regulating Valves; 5- Air Intake-Gas Freeing
D) 1- Demister; 2-O2 Analyzer; 3-Regulating Valves; 4-Air Intake-Gas Freeing; 5- Blowers

Correct Answer: B (1- Demister; 2-O2 Analyzer; 3-Air Intake-Gas Freeing; 4- Regulating Valves; 5- Blowers)

Practice Set 4

Q1. Working instructions specifying the manner in which the ship is to be loaded and ballasted to avoid unacceptable stresses in the vessel's structure is required for tankers:

A) Greater than 50 metres in length
B) Greater than 75 meters in length
C) Greater than 100 meters in length
D) Greater than 150 meters in length **Correct answer : Greater than 150 meters in length**

Correct Answer: D (Greater than 150 meters in length **Correct answer : Greater than 150 meters in length)**

Q2. A self-contained breathing apparatus is used to

A) make underwater repairs to barges
B) determine if the air in a tank is safe for men
C) enter areas that may contain dangerous fumes or lack oxygen
D) resuscitate an unconscious person **Correct answer : enter areas that may contain dangerous fumes or lack oxygen**

Correct Answer: C (enter areas that may contain dangerous fumes or lack oxygen)

Q3. Identify correct labelling for Tanker VEC

A) 1- P/V Valve; 2- P/V Breaker; 3- Mast Riser; 4 - Deck water seal
B) 1- P/V Valve; 2- P/V Breaker; 3- Deck water seal; 4- Mast Riser
C) 1- Mast Riser; 2- P/V Breaker; 3- P/V Valve; - Deck water seal
D) 1- P/V Valve; 2- Mast Riser; 3- P/V Breaker; 4 - Deck water seal

Correct Answer: D (1- P/V Valve; 2- Mast Riser; 3- P/V Breaker; 4 - Deck water seal)

Q4. A combustible gas indicator will operate correctly ONLY when the

A) hydrocarbon content of the atmosphere is less than the U.E.L.
B) atmosphere is deficient in oxygen
C) compartment to be tested is free of CO2 **Correct answer : hydrocarbon content of the atmosphere is less than the U.E.L.**

Correct Answer: A (hydrocarbon content of the atmosphere is less than the U.E.L.)

Q5. Part A of the the ISGOTT Ship/Shore Safety Check-List provides operational procedures for:

A) Transfer of bulk liquids, physical checks
B) Transfer of bulk liquids - elements to be verified verbally
C) Additional considerations for Bulk Liquid Chemicals
D) Additional considerations for Bulk Liquefied Gases **Correct answer : Transfer of bulk liquids, physical checks**

Correct Answer: A (Transfer of bulk liquids, physical checks)

Q6. The approval period for a shipboard Oil Pollution Emergency Plan expires after

A) two years
B) three years
C) four years
D) five years **Correct answer : five years**

Correct Answer: D (five years **Correct answer : five years)**

Q7. Which of the listed functions is the purpose of a gas scrubber in an inert gas generation system?

A) Cools the inert gas.
B) Maintains the oxygen content at 5% by volume.
C) Bleeds off static electricity in the inert gas.
D) Maintains flow to the water seal on the gas main. **Correct answer : Cools the inert gas.**

Correct Answer: A (Cools the inert gas.)

Q8. With an increase in temperature the volume of flammable and combustible liquids

A) expands
B) contracts
C) remains constant
D) remains constant if pressure remains constant **Correct answer : expands**

Correct Answer: A (expands)

Q9. The standard unit of liquid volume used in the petroleum industry, as well as the tanker industry, is a __________.

A) barrel
B) drum
C) gallon
D) liter **Correct answer : barrel**

Correct Answer: A (barrel)

Q10. Combustible gas indicators operate by drawing an air sample into the instrument

A) over an electrically heated platinum filament
B) where it is mixed with nitrogen
C) where it is ignited by a sparking device
D) where its specific gravity is measured **Correct answer : over an electrically heated platinum filament**

Correct Answer: A (over an electrically heated platinum filament)

Q11. Each ship having an inert gas system must have a portable instrument to measure concentrations of hydrocarbon vapor in inert atmospheres and also to measure

A) nitrogen
B) oxygen
C) carbon dioxide
D) water vapor **Correct answer : oxygen**

Correct Answer: B (oxygen)

Q12. As required by Reg.31 of Annex 1 of Marpol the ODMCS should be provided with a recording device for recording the discharge in litres per nautical mile and total quantity discharged. This discharge should be kept onboard at least for a period of

A) Intermittent, 1 year
B) Continuous, 1 year
C) Intermittent, 3 years
D) Continuous, 3 years **Correct answer : Continuous, 3 years**

Correct Answer: D (Continuous, 3 years **Correct answer : Continuous, 3 years)**

Q13. Which is the most accurate instrument for measuring the amount of oxygen in the atmosphere of a confined space?

A) Combustible gas indicator
B) Oxygen indicator
C) Flame safety lamp **Correct answer : Oxygen indicator**

Correct Answer: B (Oxygen indicator)

Q14. After each reading of an oxygen indicator, the instrument should be purged with

A) CO2
B) fresh air
C) the tested compartment's air
D) water **Correct answer : fresh air**

Correct Answer: B (fresh air)

Q15. The maneuvering vessel normally berths on which side during STS operations?

A) Any side to
B) Port side to
C) Starboard side to
D) As instructed by the mother vessel **Correct answer : Starboard side to**

Correct Answer: C (Starboard side to)

Q16. When the cofferdam to be purged to check the leakage and remove the leaked oil?

A) Before starting the cargo operation
B) Before starting the cargo operation and after the cargo operation is completed
C) During the cargo operation
D) After the cargo operation is completed **Correct answer : Before starting the cargo operation and after the cargo operation is completed**

Correct Answer: B (Before starting the cargo operation and after the cargo operation is completed)

Q17. What does "I" stand for in the centrifugal pump diagram?

A) Eye (Suction)
B) Shaft
C) Impeller
D) Diffuser **Correct answer : Shaft**

Correct Answer: B (Shaft)

Q18. The area indicated by the number "III" in the pump diagram is known as the

A) Speed Torque Control Valve
B) Cofferdam Purging
C) Local Control Valve
D) Cargo Purging Connection **Correct answer : Speed Torque Control Valve**

Correct Answer: A (Speed Torque Control Valve)

Q19. How the Framo pump has almost eliminated cavitation for free flowing liquids?

A) Volute casing is designed in such a way, that its smoothness avoids cavitation
B) The backwards curved vanes eliminate the cavitation
C) The hydrofoil design of the curved vanes, eliminate cavitation
D) Since the suction pipe is eliminated, cavitation is also eliminated for liquids with free flow **Correct answer : Since the suction pipe is eliminated, cavitation is also eliminated for liquids with free flow**

Correct Answer: D (Since the suction pipe is eliminated, cavitation is also eliminated for liquids with free flow **Corre

Q20. State if the following is True or False? After fixing the snap on hydraulic couplings of portable Framo, secure it with locking ring-only then the arrangement is safe.

A) True
B) False **Correct answer : True**

Correct Answer: A (True)

Q21. Select the correct statement regarding pressure.

A) Pressure type vacuum valves can prevent Backflow.
B) Dynamic pressure is a function of the fluid velocity and its density.
C) The water reservoir of a self-priming pump may be at the back of the impeller.
D) Stripping pumps are not of the self-priming type. **Correct answer : Dynamic pressure is a function of the fluid velocity and its density.**

Correct Answer: B (Dynamic pressure is a function of the fluid velocity and its density.)

Q22. How portable pumps are used?

A) Potable pumps are used in parallel with the main Framo pump to discharge the tank in a short time
B) When main pump is not working, portable pumps are used to discharge the liquid cargo to the terminal
C) Potable pumps are used in series with the main Framo pump to discharge the tank with large pressure.
D) When main pump is not working, the portable pump is used to transfer the cargo from this tank to another empty tank, where the main pump is working **Correct answer : When main pump is not working, the portable pump is used to transfer the cargo from this tank to another empty tank, where the main pump is working**

Correct Answer: D (When main pump is not working, the portable pump is used to transfer the cargo from this tank to

Q23. What does "II" indicate in the pump diagram?

A) Eye (Suction)
B) Shaft
C) Impeller
D) Diffuser **Correct answer : Eye (Suction)**

Correct Answer: A (Eye (Suction))

Q24. Which number represents the Hydraulic Motor in the pump cross-section illustration?

A) I
B) II
C) III
D) IV **Correct answer : I**

Correct Answer: A (I)

Q25. The part "IX" in the centrifugal pump diagram indicates

A) Stuffing Box
B) Shaft Sleeve
C) Packing
D) Casing **Correct answer : Packing**

Correct Answer: C (Packing)

Q26. Why a two way valve is provided in the air or inert gas line provided to check the leakage of oil in the cofferdam?

A) One is for the normally closed position and the other is for checking the leakage of hydraulic oil in the motor side
B) One is for the normally closed position and the other is for checking the leakage of cargo oil in the pump side
C) One is for the normally closed position, the other is for checking the leakage of cargo oil in the pump side and the third one is for checking the leakage of hydraulic oil in the motor side **Correct answer : One is for the normally closed position, the other is for checking the leakage of cargo oil in the pump side and the third one is for checking the leakage of hydraulic oil in the motor side**

Correct Answer: C (One is for the normally closed position, the other is for checking the leakage of cargo oil in the pu

Q27. From the pump illustration, identify the part marked - "IV".

A) Hydraulic Motor
B) Volute Casing
C) Impeller
D) Anti-rotation Brake **Correct answer : Volute Casing**

Correct Answer: B (Volute Casing)

Q28. How vacuum drain is done in Framo pumps?

A) Pump has a different vacuum line which is connected to the cargo tank well; vacuum pump is used to suck the oil out to make the well dry
B) Vacuum is created in the stripping line to pump the cargo out to make the well dry
C) Vacuum drain is used to check leakage in pump's cofferdam
D) vacuum drain is part of the pump casing **Correct answer : Pump has a different vacuum line which is connected to the cargo tank well; vacuum pump is used to suck the oil out to make the well dry**

Correct Answer: A (Pump has a different vacuum line which is connected to the cargo tank well; vacuum pump is use

Q29. The number "10" in the Tanker cargo pump assembly indicates a __________.

A) Top Plate
B) Stripping Valve
C) Resilient Mounting **Correct answer : Stripping Valve**

Correct Answer: B (Stripping Valve)

Q30. Why the cargo pump in the Framo cannot be overloaded?

A) The pressure of hydraulic oil to the motor driving the cargo pump is designed in such a way that, the flow is always kept constant
B) The pressure of hydraulic oil to the motor driving the cargo pump is designed in such a way that, the pump runs at the optimum speed
C) The pressure of hydraulic oil to the motor driving the cargo pump is designed in such a way that, when the pump over speeds, the flow of oil is reduced
D) The pressure of hydraulic oil to the motor driving the cargo pump is designed in such a way that, the pressure can run the pump at 75% of the capacity **Correct answer : The pressure of hydraulic oil to the motor driving the cargo pump is designed in such a way that, the pump runs at the optimum speed**

Correct Answer: B (The pressure of hydraulic oil to the motor driving the cargo pump is designed in such a way that,

Q31. What does figure "V" represent in the centrifugal discharge diagram?

A) Vane
B) Eye of impeller
C) Discharge nozzle
D) None of the above **Correct answer : Discharge nozzle**

Correct Answer: C (Discharge nozzle)

Q32. Normally how much hydraulic oil is required to run the Framo portable pump?

A) 360 litres per min
B) 250 litres per min
C) 140 litres per min
D) 90 litres per min **Correct answer : 140 litres per min**

Correct Answer: C (140 litres per min)

Q33. Select the correct statement regarding operation of the portable pumps?

A) Before using portable pump, rig the pump on the deck and while pumping, lower the hose in the tank
B) When the liquid cargo level goes down, lower the hose accordingly, keeping the rigged pump on the deck
C) Make sure that the portable pump is completely submerged and lowered till the bottom of the tank and then start the pumping operation
D) Ensure that only the lower half of the pump is submerged and is lowered as the cargo level falls. **Correct answer : Ensure that only the lower half of the pump is submerged and is lowered as the cargo level falls.**

Correct Answer: D (Ensure that only the lower half of the pump is submerged and is lowered as the cargo level falls.

Q34. What is the pressure of the hydraulic oil in the return side after running the hydraulic motor?

A) Around 3 Bar
B) Around 10 bar
C) Around 15 bar
D) Above 15 bar **Correct answer : Around 3 Bar**

Correct Answer: A (Around 3 Bar)

Q35. What does "V" indicate in the pump illustration?

A) Speed Torque Control Valve
B) Cofferdam Purging
C) Local Control Valve
D) Cofferdam Check Pipe **Correct answer : Cofferdam Purging**

Correct Answer: B (Cofferdam Purging)

Q36. What does figure "VII" represent in the pump assembly?

A) Hydraulic Motor
B) Sleeve
C) Wear rings
D) Mechanical Oil Seal **Correct answer : Mechanical Oil Seal**

Correct Answer: D (Mechanical Oil Seal **Correct answer : Mechanical Oil Seal)**

Q37. Normally what is the discharge capacity of Framo portable pumps?

A) 150 cubic meter per hour
B) 100 cubic meter per hour
C) 70 cubic meter per hour
D) 200 cubic meter per hour **Correct answer : 70 cubic meter per hour**

Correct Answer: C (70 cubic meter per hour)

Q38. What is the function of the item marked "II" in the pump diagram?

A) Impeller
B) Hydraulic Motor
C) Anti-rotation Brake
D) Suction Well **Correct answer : Anti-rotation Brake**

Correct Answer: C (Anti-rotation Brake)

Q39. In a centrifugal pump:

A) The discharge side has a negative pressure.
B) The fluid moves in a direction opposite to the impeller.
C) The fluid moves in the same direction as the impeller.
D) The width of the discharge pipeline decreases outwards. **Correct answer : The fluid moves in the same direction as the impeller.**

Correct Answer: C (The fluid moves in the same direction as the impeller.)

Q40. What does "V" indicate in the below illustration?

A) Bearings
B) Mechanical Oil Seal
C) Suction Well
D) Sleeve **Correct answer : Suction Well**

Correct Answer: C (Suction Well)

Q41. What does "VIII" indicate in the centrifugal pump diagram?

A) Impeller
B) Discharge Nozzle
C) Eye of Impeller
D) None of the above **Correct answer : Eye of Impeller**

Correct Answer: C (Eye of Impeller)

Q42. How purging and seal monitoring is done?

A) Air or inert gas pipe connected to the line, the two way valve which remains closed is moved to hydraulic oil side and air or inert gas is opened and if there is any leak, it is purged out and the quantity is measured.
B) Air or inert gas pipe connected to the line, the two way valve is moved to hydraulic oil side and air or inert gas is opened and if there is any leak, it is purged out and the quantity is measured, then the two way valve is moved to the cargo oil side and the air or inert gas is opened and if there is any leak, it is purged out and the quantity is measured
C) Air or inert gas pipe connected to the line, the two way valve is moved to cargo oil side and air or inert gas is opened and if there is any leak, it is purged out and the quantity is measured. **Correct answer : Air or inert gas pipe connected to the line, the two way valve is moved to hydraulic oil side and air or inert gas is opened and if there is any leak, it is purged out and the quantity is measured, then the two way valve is moved to the cargo oil side and the air or inert gas is opened and if there is any leak, it is purged out and the quantity is measured**

Correct Answer: B (Air or inert gas pipe connected to the line, the two way valve is moved to hydraulic oil side and ai

Q43. Bernoulli's principle states that:

A) pressure (P) x height (H) is a constant
B) height (H) x velocity (V) is a constant
C) pressure (P) x velocity (V) is a constant
D) pressure (P) x density (D) is a constant **Correct answer : pressure (P) x velocity (V) is a constant**

Correct Answer: C (pressure (P) x velocity (V) is a constant)

Q44. From the below illustration, identify the name of the marking "14".

A) Resilient Mounting
B) Deck Trunk
C) Cargo Pipe
D) Hydraulic Pressure Pipe **Correct answer : Cargo Pipe**

Correct Answer: C (Cargo Pipe)

Q45. From the below illustration, identify the name of the marking "V".

A) Diffuser
B) Shaft
C) Eye (Suction)
D) Impeller **Correct answer : Diffuser**

Correct Answer: A (Diffuser)

Q46. How the speed of the portable pump can be varied?

A) Portable pump speed cannot be varied
B) Using the flow control valve of the main cargo pump
C) By varying the speed of the feed pump
D) The portable pump is always rigged with its own flow control valve which is used for speed variation **Correct answer : The portable pump is always rigged with its own flow control valve which is used for speed variation**

Correct Answer: D (The portable pump is always rigged with its own flow control valve which is used for speed variat

Q47. What kind of pumps can move extremely thick fluids like sludge without clogging?

A) Rotary vane pumps
B) Reciprocating pumps
C) Deep well pumps
D) Screw pumps **Correct answer : Screw pumps**

Correct Answer: D (Screw pumps **Correct answer : Screw pumps)**

Q48. What is the pressure of the hydraulic oil in the inlet side for running the hydraulic motor?

A) Around 50 Bar
B) Around 170 bar
C) Around 100 bar
D) Around 25 bar **Correct answer : Around 170 bar**

Correct Answer: B (Around 170 bar)

Q49. The reciprocating pump creates a suction effect when there is a

A) decrease in pressure
B) increase in pressure
C) no change in pressure
D) decrease or an increase in pressure **Correct answer : decrease in pressure**

Correct Answer: A (decrease in pressure)

Q50. What is represented by the figure "VI" in the pump assembly illustration?

A) Eye of impeller
B) Shaft
C) Impeller
D) Casing **Correct answer : Impeller**

Correct Answer: C (Impeller)